

Single-User Feedback: From Broadcast Erasure Channels Toward General Broadcast Models

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Abstract

This project studies secure communication, with a focus on the two-user broadcast packet erasure channel under single-user feedback. Starting from the information-theoretic secrecy viewpoint, the report reviews the wiretap channel, broadcast erasure channels with feedback, secure broadcast erasure communication, and feedback in more general broadcast-channel models. The main focus is the single-user feedback setting, where only one receiver reports its channel state to the transmitter while the other receiver provides no feedback. This reduced-feedback model is important because it lowers feedback overhead, but also creates an asymmetric secrecy problem.

The report examines the three-phase achievable scheme for this model. In the first phase, random packets are used to generate a secret key for the feedback receiver and a randomizer for the non-feedback receiver. In the second phase, wiretap-channel coset coding protects the non-feedback receiver's message. In the third phase, ARQ-based opportunistic network coding delivers the feedback receiver's encrypted message while creating useful side information for the non-feedback receiver. Numerical simulations are then used to illustrate the achievable secure rate region and compare it with full-feedback and no-feedback baselines. The results show that single-user feedback does not fully recover the full-feedback secure capacity region, but it still provides a substantial gain over the no-feedback case. In particular, the non-feedback receiver can maintain a positive secure rate over a meaningful range even when its channel is weaker. Finally, the report discusses why extending this structure to general noisy broadcast channels is difficult, since the clear received-or-erased side-information structure of BPECs no longer directly applies.

1 Introduction

Wireless communication is inherently broadcast in nature. The broadcast channel was introduced as a fundamental multi-user communication model in which one transmitter communicates with multiple receivers over a shared channel [1]. Unlike a point-to-point wired link, a wireless transmission may be observed by multiple receivers at the same time, including receivers for whom the message is not intended. This property makes broadcast communication useful for serving multiple users, but it also introduces two fundamental difficulties. First, the transmitter must manage different channel conditions across receivers. Second, when private messages are sent over the same broadcast medium, the system must guarantee reliable decoding at the intended receiver and further confidentiality against unintended receivers.

Information-theoretic security provides a useful framework for studying this problem. Starting from Wyner's wiretap channel, secrecy can be formulated in terms of the uncertainty that remains at an eavesdropper after observing the channel output [2]. Instead of depending on computational hardness or on hiding the coding scheme, it is obtained from the statistical structure of the communication channel itself. This idea becomes especially interesting in broadcast erasure channels, where different receivers may observe different subsets of transmitted packets due to random erasures.

Feedback further changes the role of the channel. In a degraded broadcast channel, feedback does not increase capacity [3]. However, in general broadcast channels, feedback can enlarge the achievable rate region by allowing the transmitter to adapt future transmissions according to past receiver observations [4]. In broadcast packet erasure channels, feedback is particularly useful because the transmitter can learn which packets were received or erased at each receiver. This state information can be used for retransmission, network coding, and, in secure communication, secret-key generation. Prior work on broadcast erasure channels with full state feedback shows that feedback can support secure communication by using random erasures to create secret keys and by using those keys to protect private messages [5].

However, requiring every receiver to provide feedback may be costly in practical systems. Feedback consumes reverse-link resources and may introduce additional overhead, especially when the number of receivers grows. This motivates the study of reduced-feedback models. Therefore, the single-user feedback setting asks whether secure communication is still possible when only one receiver reports its channel state to the transmitter. The main contribution is deriving the feedback receiver can benefit from its own feedback and the non-feedback receiver can obtain a positive secure rate from the feedback provided by another user [6].

This report studies this question through the two-user broadcast packet erasure channel with single-user feedback and review the related foundations. Moreover, to review the theoretical model, numerical simulations in order to illustrate the behavior of the achievable region is also provided, evaluating the secure

rate tradeoff between the feedback receiver and the non-feedback receiver under different erasure probabilities.

Finally, the report discusses why extending the single-user feedback scheme beyond erasure channels is nontrivial. In a packet erasure model, the feedback state directly identifies which packets are available as side information at the receivers. This clear received-or-erased structure supports key generation, randomizer construction, and network-coded retransmission. In a general noisy broadcast channel, receiver observations are no longer simple erasure patterns, so it becomes less direct to define equivalent secrecy resources or opportunistic coding operations. Therefore, the project uses the BPEC setting as a structured foundation and then identifies the main obstacles toward extending single-user secure feedback ideas to more general broadcast models.

2 Background and Related Work

2.1 Wiretap Channel and Information-Theoretic Secrecy

The wiretap channel considers a sender communicates with a legitimate receiver through a main channel, while in Wyner's degraded wiretap channel, the wiretapper observes the main-channel output through a second noisy channel [2]. More specifically, the encoder maps a source block into a channel input sequence X^N , the legitimate receiver observes the main channel output Y^N , and the eavesdropper observes Z^N through an additional wiretap channel. This yields the degraded structure $X \rightarrow Y \rightarrow Z$, meaning that the eavesdropper has access only to a noisier observation than the intended receiver.

The key issue in this model is not only whether the legitimate receiver can decode reliably, but also how much uncertainty remains at the eavesdropper. [2] measured secrecy by the equivocation

$$\Delta = \frac{1}{K} H(S^K | Z^N),$$

where the normalization by K represents the equivocation per source symbol. Represents the remaining uncertainty of the source message after the wiretapper has observed the channel output. A larger equivocation indicates stronger secrecy. In particular, when the equivocation approaches the source entropy H_s , the wiretapper essentially learns nothing useful about the message.

An important operational implication of this formulation is that secrecy is not obtained by hiding the coding scheme itself. In Wyner's model, the codebooks and the encoder-decoder structure are assumed to be known to the wiretapper. Therefore, the protection does not come from computational difficulty or from a secret codebook, but from the statistical difference between the legitimate receiver's observation and the wiretapper's observation. The encoder can introduce controlled randomness so that the legitimate receiver

can still decode the intended message reliably, while the wiretapper's observation remains insufficient to identify the source message. Therefore, the wiretap channel provides a physical-layer notion of security, which is the channel itself becomes part of the security mechanism.

Based on the definition, communication performance is characterized by achievable (R, d) pairs, where R denotes the transmission rate and d denotes the achievable equivocation level. A pair (R, d) is achievable if there exists an encoding and decoding scheme such that the transmission rate is close to R , the equivocation is close to d , and the probability of decoding error in the legitimate receiver can be made arbitrarily small. This formulation is important because it expresses security as a tradeoff between rate and information leakage, rather than as a purely cryptographic notion based on hidden keys.

The significance of the model extends beyond the original point-to-point setting. Once secrecy is formulated in terms of achievable rate-equivocation pairs, one can further ask for the maximum reliable rate under perfect secrecy, which leads to the notion of secrecy capacity. In more general multi-user settings, the same viewpoint naturally extends to secrecy regions, where multiple rates and secrecy constraints must be characterized simultaneously [7]. For this reason, the wiretap channel is widely regarded as the conceptual starting point of modern information-theoretic security. This broader line of development is summarized in the survey [8], which reviews how the wiretap model extends to a general framework for information-theoretic security, including secrecy capacity, feedback effects, and more general network settings.

This viewpoint also explains why feedback and erasures become useful resources in later broadcast-erasure models. In the original wiretap setting without feedback, secure communication mainly relies on the legitimate receiver having a better observation than the eavesdropper. However, when public feedback is introduced, the transmitter can learn which packets were received or erased at different terminals. This channel-state information can be used to create shared randomness or secret keys, and later to protect confidential messages. Therefore, it can also create new secrecy resources that are absent in the classical no-feedback wiretap channel.

2.2 Broadcast Packet Erasure Channels with Feedback

A broadcast packet erasure channel (BPEC) serves as an abstraction for wireless communication [9]. Since the channel is broadcast by nature, one transmission may be received by multiple users at the same time. In addition, due to fading, interference, or collision, the received packet may either arrive correctly or be erased. Therefore, broadcast packet erasure channel is widely applied to construct lossy wireless networks. In the 1-to-K setting, one source transmits to K receivers, and in each channel use different subsets of receivers may successfully obtain the transmitted packet.

The model is expanded further in [10] by introducing the feedback, the source has instant channel output feedback after each transmission. The transmitter can adapt future transmissions according to the users who have received earlier packets successfully. [10] showed the capacity region for the general 1-to-3 broadcast packet erasure channels with channel output feedback, and also characterized two important classes of 1-to- K channels, namely symmetric channels and spatially independent channels with one-sided fairness constraints. For more general cases, the paper provides outer and inner bounds that can be evaluated numerically, which shows that feedback plays a central role in understanding the achievable rate region. Related work [11] also investigates the multiuser broadcast erasure channel with feedback. The paper derives a generic outer bound, proposes queue-based feedback coding algorithms, and proves capacity-achieving results for several channel settings. These non-secure studies provide the theoretical background for later discussion of secure broadcast erasure channels.

2.3 Secret Communication over BPEC with Feedback

[5] studies secure communication over a 1-to- K broadcast erasure channel with strictly causal and public state-feedback, where the sender has one private message for each receiver and each message must remain information-theoretically secret from all the other receivers. Compared with the classical wiretap setting, this model is more involved because secrecy must be guaranteed simultaneously against multiple unintended receivers who may also collude. At the same time, compared with the ordinary broadcast erasure channel with feedback, the sender can no longer treat overheard side information as purely beneficial, since that same side information may leak confidential information to other users.

A particularly important feature is that it makes the relation between the secure and non-secure problems explicit. Rather than presenting secrecy as an unrelated extra constraint, the authors show that the secure capacity expression can be understood as the non-secure communication structure together with an additional secrecy overhead. For honest-but-curious users, the resulting secure capacity characterization is

$$\max_{j \in \{1, \dots, K\}} \frac{R_j \left(1 - \frac{\prod_{k=1}^K \delta_k}{\delta_j}\right)}{(1 - \delta_j) \frac{\prod_{k=1}^K \delta_k}{\delta_j} \left(1 - \prod_{k=1}^K \delta_k\right)} + \max_{\pi} \sum_{i=1}^K \frac{R_{\pi_i}}{1 - \prod_{k=1}^i \delta_{\pi_k}} \leq 1. \quad (1)$$

where the second term corresponds to the underlying non-secure broadcast transmission structure, while the first term captures the extra communication cost required by secrecy. This characterization identifies the security penalty as a precise rate cost that can be quantified within the same communication framework.

One of the central ideas of the paper is its two-phase achievability scheme. In the first phase, the sender exploits random packet erasures together with public state feedback to establish pairwise secret keys with

each receiver. In the second phase, these keys are used to protect the private messages. This two-phase structure is important because it explains how secrecy can be built from the channel randomness itself rather than being assumed in advance.

The key generation phase relies on the fact that, due to random erasures, different receivers do not observe the same packets [12, 13]. Some transmitted random packets are received by one receiver but erased at the others. With feedback, the sender can identify these receiver-specific observations and turn them into secret keys shared only with the intended receiver. In the other words, erasures create asymmetry across receivers, which becomes the source of secrecy [14].

The required amount of key is not necessarily equal to the full message size. Instead, it is determined by how much encrypted information the unintended receivers might jointly collect. This makes the scheme more efficient than a naive one-time-pad approach in which every message symbol would require an equal amount of secret key in advance. It also clarifies the secrecy cost of the problem, showing that the system only needs enough key material to cover the information that may actually leak through the broadcast process.

The paper presents an achievable coding scheme, while its converse argument further explains why the scheme is structurally tied to the problem itself. In particular, a converse proof is developed that reflects the same two-phase logic as the achievability result. At a high level, this converse does more than provide an outer bound, showing that secure communication is fundamentally constrained by the balance between keys generation and consumption. In other words, a scheme cannot sustain an arbitrary secure message rate unless the channel and feedback mechanism can generate secrecy resources fast enough to support the encrypted transmission phase. This is an important point, because it suggests that the key-generation stage and the encrypted transmission stage are closely related to the basic limit of secure communication in this model rather than an arbitrary construction. Therefore, the two-phase structure should be understood not just as an intuitive coding idea, but as a reflection of the underlying resource constraint of the problem itself.

This interpretation also explains why the secure problem is strictly more demanding than the corresponding non-secure broadcast erasure problem. In the non-secure setting, feedback is used mainly to improve transmission efficiency by exploiting receiver side information. However, the same channel resources must support both reliable delivery and confidentiality in the secure setting, . This additional requirement appears directly in the secure capacity formula: as the paper points out, in Theorem 3 the first term can be interpreted as the overhead for security, and the duration of the key generation phase is proportional to this term. Therefore, the secure rate region should be understood not simply as a mathematical boundary, but as a precise statement of the tradeoff between communication efficiency and secrecy creation in a multi-user broadcast system.

2.4 General Broadcast Channels with Feedback

The previous subsections focus on broadcast packet erasure channels, where each receiver either obtains the transmitted packet correctly or observes an erasure. This model is analytically convenient because the feedback reveals a clear channel state. The transmitter can identify which receivers have received each packet and which receivers have not. However, broadcast communication is not limited to erasure models. In a general discrete memoryless broadcast channel, the transmitter sends an input symbol X , and the two receivers observe channel outputs Y_1 and Y_2 according to a transition probability $P_{Y_1, Y_2 | X}$. With feedback, the channel input at time t may depend not only on the messages but also on the past channel outputs, such as Y_1^{t-1} and Y_2^{t-1} .

The role of feedback in general broadcast channels is more subtle than in point-to-point channels. For a point-to-point memoryless channel, feedback does not increase capacity. For broadcast channels, however, feedback can enlarge the achievable rate region in general. A classical example due to Dueck [4] shows that feedback can strictly improve the capacity region of a broadcast channel. This result indicates that feedback change how information is coordinated across multiple receivers.

One way to understand this gain is through the idea of resolving uncertainty created in previous transmissions. In the two-receiver discrete memoryless broadcast channel with feedback, Venkataramanan [15] proposed an achievable rate region based on block-Markov superposition coding. Their scheme starts from Marton's coding idea [16] for the no-feedback broadcast channel. When the receivers cannot uniquely decode their intended messages at the end of one block, each receiver is left with a list of possible codewords. Through feedback, the transmitter learns information about these lists and sends resolution information in the next block. The key observation is that the resolution information needed by the two receivers is generally correlated, so it can be transmitted more efficiently than two independent pieces of information.

This viewpoint is conceptually related to the role of feedback in broadcast erasure channels. In both cases, feedback allows the transmitter to use past receiver observations to shape future transmissions. In the erasure case, this often appears as packet retransmission, network coding, or secret-key generation based on who received which packets. In the general broadcast channel, the same idea becomes less explicit. The transmitter does not simply learn an erasure pattern, but rather uses past channel outputs to help resolve the receivers' decoding uncertainty. Therefore, the erasure model can be viewed as a structured setting in which the benefit of feedback becomes especially transparent.

At the same time, this comparison also shows why extending single-user feedback schemes from BPECs to general broadcast channels is difficult. In a BPEC, the feedback state directly identifies side information at the receivers and allows the transmitter to construct keys, randomizers, and network-coded packets with clear

algebraic structure. In a general noisy broadcast channel, receiver observations are not simply received-or-erased packets. Thus, it is no longer immediate how to define the side information available at each receiver, how to extract secrecy resources from it, or how to design an equivalent opportunistic coding step. For this reason, general broadcast-channel feedback results provide an important conceptual guide, but they do not directly yield a single-user secure feedback scheme for general channels.

3 Single-User Feedback over BPEC

Previous works show that return state information to the transmitters is able to achieve a higher achievable secure rate. While most of the classical model focus on fully feedback model, indicating that each of the user is required to feedback, the considerable overhead that introduced by feedback system result in the limitation of reverse-link resources.

The restriction motivates the design of a single-user feedback model, where only one user provides the feedback state to the transmitter. Investigating the reduced model, discussing the plausibility of asymmetric structure, and analyzing the advantage of the model. The purpose of this section is to examine the conceptual and analytical foundation of the single-user feedback model. First, the framework will be introduced in the following chapter. Furthermore, the methodology and achievable rate region will also be revealed.

3.1 System Model and Feedback Framework

3.1.1 Channel Model and Feedback

To simplify the channel model, a packet erasure channel is used to characterized the state dependent Gaussian channel based on [5]. Furthermore, the system consider two-user broadcast packet erasure channel (BPEC) consisting of one transmitter and two receivers, denoted as R_{X_1} and R_{X_2} respectively. At each time index t , the transmitter sends one packet $X[t] \in \mathbb{F}_q$, and the corresponding channel outputs at the two receivers are denoted by $Y_1[t]$ and $Y_2[t]$, respectively.

Note that R_{X_1} is the feedback receiver which will send its channel state back to the transmitter T_x . After each transmission, it reveals its channel state to the transmitter with unit delay. This feedback is public in the sense that R_{X_2} can also overhear it. Hence, at time t , both T_x and R_{X_2} have access to the past feedback sequence $S_1[1 : t - 1]$, while R_{X_2} does not provide its own channel state feedback. The two confidential messages intended for R_{X_1} and R_{X_2} are denoted by M_1 and M_2 , respectively.

The channel is memoryless and follows the joint channel law

$$P_{Y_1, Y_2|X}(y_1, y_2|x) = \begin{cases} \delta_1 - \delta_{1,2}, & y_1 = \star, y_2 = x \\ \delta_2 - \delta_{1,2}, & y_1 = x, y_2 = \star \\ \delta_{1,2}, & y_1 = \star, y_2 = \star \\ 1 - \delta_1 - \delta_2 + \delta_{1,2}, & y_1 = x, y_2 = x \end{cases} \quad (2)$$

where $\star$ is "erasure symbol", δ_i is the probability of erasure at R_{x_i} . Therefore, the parameters must satisfy

$$\max\{0, \delta_1 + \delta_2 - 1\} \leq \delta_{1,2} \leq \min\{\delta_1, \delta_2\}, \quad (3)$$

so that all four transition probabilities in (2) are nonnegative.

For $i = 1, 2$, the channel state at time t is defined as

$$S_i[t] = \mathbf{1}\{Y_i[t] \neq \star\}, \quad (4)$$

which indicates whether the packet is successfully received by R_{x_i} . Since only $S_1[t]$ is fed back, the transmitter can adapt its future transmissions according to the reception history of R_{X_1} , but not according to the private reception history of R_{X_2} .

3.1.2 Security and Reliability Constraint

In order to evaluate the system, several assumptions are set. First, the error rate must decrease to zero as the blocklength n approaches infinite. The criteria ensure that the receivers decoding the message successfully, which guarantees the reliability of the system. However, it also results in the limitation of the achievable rate region.

$$\lim_{n \rightarrow \infty} \Pr\{\hat{M}_i \neq M_i\} = 0, \quad i = 1, 2; \quad (5)$$

Security is another concern, which requires the guaranty that an unintended receiver can only obtain negligible noise-like information.

$$\lim_{n \rightarrow \infty} I(M_1; Y_2[1 : n], S_1[1 : n]) = 0, \quad (6)$$

$$\lim_{n \rightarrow \infty} I(M_2; Y_1[1 : n]) = 0. \quad (7)$$

Therefore, the criteria are given to satisfy the security constraint. The first condition indicates that the information observed by R_{X_2} , including both $Y_2[1 : n]$ and the public feedback sequence $S_1[1 : n]$, is asymptotically independent of M_1 . Similarly, the second condition indicates that the information observed by R_{X_1} is asymptotically independent of M_2 . This condition guarantees information-theoretic secrecy, independent of computational assumptions. Intuitively, part of the transmission must be used to generate shared randomness or secret keys, which consume channel resources.

3.1.3 Three-phase Transmission Structure

According to [6], single-user feedback model includes three phases, serving a different functional objective to achieve both reliability and theoretic secrecy. Additionally, the total blocklength n is the sum of the blocklength of each phase, that is,

$$n = n^{(1)} + n^{(2)} + n^{(3)}$$

where $n^{(j)}$ denotes the channel use of Phase j .

In Phase 1, T_x sends a sequence of *i.i.d* uniform distributed random symbols, generated by private random variable. With the feedback from R_{X_1} , T_x can distinguish whether each symbol is received by R_{X_1} . The secret key K_1 is then produced by the symbols successfully received by R_{X_1} processed through linear privacy amplification. In contrast, the symbols that are not received by R_{X_1} are selected to form a randomizer vector U_2 , to protect the message for R_{X_2} .

In Phase 2, the transmitter encodes the message M_2 using linear wiretap-channel (WTC) coset coding. The randomizer U_2 is used to combine with M_2 to generate the encoded vector. Since U_2 is picked from the random symbols that R_{X_1} is blind to, the confidentiality of R_{X_2} messages can be promised by properly design the length of the randomizer and the code structure. At the end of Phase 2, based on the feedback from R_{X_1} , the transmitter can identify $X_{rx1}^{(2)}$, which denotes the sub-vector of Phase 2 symbols received by R_{X_1} .

In Phase 3, the encrypted codeword W_1 for User 1 is superimposed by $Z^{(3)}$, which is a linear combination

of symbols transmitted in $X_{rx1}^{(2)}$. More precisely, the transmitter first generates

$$Z^{(3)} = X_{rx1}^{(2)} h_{\text{re_enc}},$$

and then delivers W_1 and $Z^{(3)}$ using an ARQ-type opportunistic network coding protocol. The ARQ rule is applied to the W_1 queue, that is, the transmitter moves to the next symbol of W_1 only when R_{X_1} successfully receives the current one. In contrast, the queue $Z^{(3)}$ continuously moves forward regardless of feedback. Each transmitted symbol is the superposition of the current symbols from these two queues.

Moreover, $Z^{(3)}$ does not interfere with the decoding process of R_{X_1} , because $Z^{(3)}$ is generated from $X_{rx1}^{(2)}$, which is already side information at R_{X_1} . Therefore, this part does not introduce additional secrecy leakage of M_2 to R_{X_1} . However, the W_1 symbols overheard by R_{X_2} still need to be protected by the secret key K_1 .

Note that R_{x_2} does not provide feedback, and therefore the transmitter does not directly know which packets have been received by R_{x_2} . As a result, the decodability of M_2 is not guaranteed through an explicit secret resource shared between T_x and R_{x_2} . Instead, reliable decoding relies on the fact that once R_{x_2} obtains sufficient side information, the error probability can be arbitrarily small.

3.2 Coding Methodology and Achievable Secure Rate

3.2.1 Coding Methodology

The coding methodology can be interpreted as the combination of three main tools, each addressing a different difficulty in the single-user feedback setting. First, linear privacy amplification is used in Phase 1 to extract a secret key K_1 from the random symbols received by R_{X_1} while keeping the key secret from R_{X_2} . This key is then used to encrypt the message M_1 for the feedback receiver.

Additionally, linear wiretap-channel coset coding is used to protect the message M_2 for the non-feedback receiver. The randomizer U_2 , which is formed from the Phase 1 random symbols erased at R_{X_1} , is combined with M_2 in the coset encoder. Since R_{X_1} does not observe these randomizer symbols, they provide the randomness needed to limit the information leakage of M_2 .

Opportunistic network coding with an ARQ-type structure is used in Phase 3. This phase delivers the encrypted codeword W_1 to R_{X_1} while simultaneously sending linear combinations of the symbols that R_{X_1} overheard in Phase 2. The role of this step is not merely retransmission; it also creates useful side information for R_{X_2} and improves the decodability of the non-feedback user's message. Hence, the feedback from only one receiver is used for three purposes: secret-key generation, randomizer construction, and opportunistic network coding.

3.2.2 Achievable Secure Rate

Let the confidential messages for the two receivers be M_1 and M_2 , with rates R_1 and R_2 , respectively. The objective is to design a coding scheme such that both receivers can reliably decode their intended messages while preventing any unintended receiver from obtaining non-negligible information about the confidential messages.

Under the 3-phase scheme, an achievable secure rate is determined by reliability constraints and security constraints. Furthermore, the assumptions introduced by the transmission protocol lead to the bound of the achievable region as well. Theorem 1 in [6] states that any nonnegative rate pair satisfying the following constraints is achievable:

$$R_1 \frac{1}{1 - \delta_{1,2}} + R_2 \left(\frac{1}{\delta_1 - \delta_{1,2}} + \frac{\delta_1(\delta_2 - \delta_{1,2})}{(\delta_1 - \delta_{1,2})(1 - \delta_2)} \right) \leq 1, \quad (8)$$

$$R_1 \left(\frac{1}{1 - \delta_{1,2}} + \frac{1 - \delta_2}{(1 - \delta_{1,2})(\delta_2 - \delta_{1,2})} \right) + R_2 \frac{\delta_1(1 - \delta_{1,2})}{(\delta_1 - \delta_{1,2})(1 - \delta_2)} \leq 1, \quad (9)$$

$$R_1 \left(\frac{1}{1 - \delta_1} + \frac{1 - \delta_2}{(1 - \delta_{1,2})(\delta_2 - \delta_{1,2})} \right) + R_2 \frac{\delta_1}{\delta_1 - \delta_{1,2}} \leq 1. \quad (10)$$

where δ_1 , δ_2 and $\delta_{1,2}$ denote the erasure probabilities of User 1, User 2, and their joint erasure probability, respectively.

The result shows that even though R_{X_2} does not provide feedback, the feedback from R_{X_1} can still help R_{X_2} achieve a strictly positive secure communication rate under suitable channel parameters. This is because the single-user feedback is used not only for the feedback user's secret-key generation, but also for constructing the randomizer of the non-feedback user and for enabling opportunistic network coding. Therefore, the achievable region is generally asymmetric. The feedback user benefits directly from its own feedback, while the non-feedback user benefits indirectly through the structure created by the other user's feedback.

It should be emphasized that the region in (8)–(10) is an achievable rate region of the proposed scheme, not a complete capacity characterization of the single-user feedback setting. Compared with the no-feedback case, the region demonstrates a strict gain from single-user feedback. However, compared with the full-feedback setting, the achievable region is generally smaller because only one receiver provides channel state feedback.

4 Numerical Evaluation and Simulation

This section provides a numerical evaluation of the secure rate regions discussed in the previous sections. The main objective is to translate the theoretical rate constraints into graphical regions in the (R_1, R_2) plane, so that the effect of feedback and secrecy constraints can be observed more clearly. In particular, the numerical study focuses on three questions: how the single-user feedback achievable region behaves, how it compares with full-feedback and no-feedback baselines, and how the achievable secure rate of the non-feedback receiver changes when the erasure probabilities vary.

Throughout this section, R_{X_1} denotes the feedback receiver and R_{X_2} denotes the non-feedback receiver. The erasure probabilities of the two receivers are denoted by δ_1 and δ_2 , and the joint erasure probability is denoted by $\delta_{1,2}$. Unless otherwise specified, independent erasures are assumed, so that

$$\delta_{1,2} = \delta_1 \delta_2.$$

This assumption makes the numerical comparisons easier to interpret because the joint erasure probability is determined by the two marginal erasure probabilities.

4.1 Rate Region

For the representative channel setting,

$$\delta_1 = 0.4, \quad \delta_2 = 0.6, \quad \delta_{1,2} = 0.24,$$

the achievable secure rate region is plotted in Fig. 1. Since

$$\delta_{1,2} = \delta_1 \delta_2,$$

this setting corresponds to the independent erasure case. Also, due to $\delta_2 > \delta_1$, R_{X_2} has the weaker channel. This choice is useful for illustrating the main qualitative feature of the single-user feedback scheme, which is even when the non-feedback receiver has a worse channel condition, it can still achieve a positive secure rate with the help of the feedback receiver's state information.

For each fixed R_1 , the numerical evaluation computes the largest feasible value of R_2 allowed by the three single-user feedback rate inequalities, that is (8)–(10). The three upper bounds on R_2 induced by these inequalities equivalently. The shaded area below this boundary represents the set of rate pairs satisfying all the constraints.

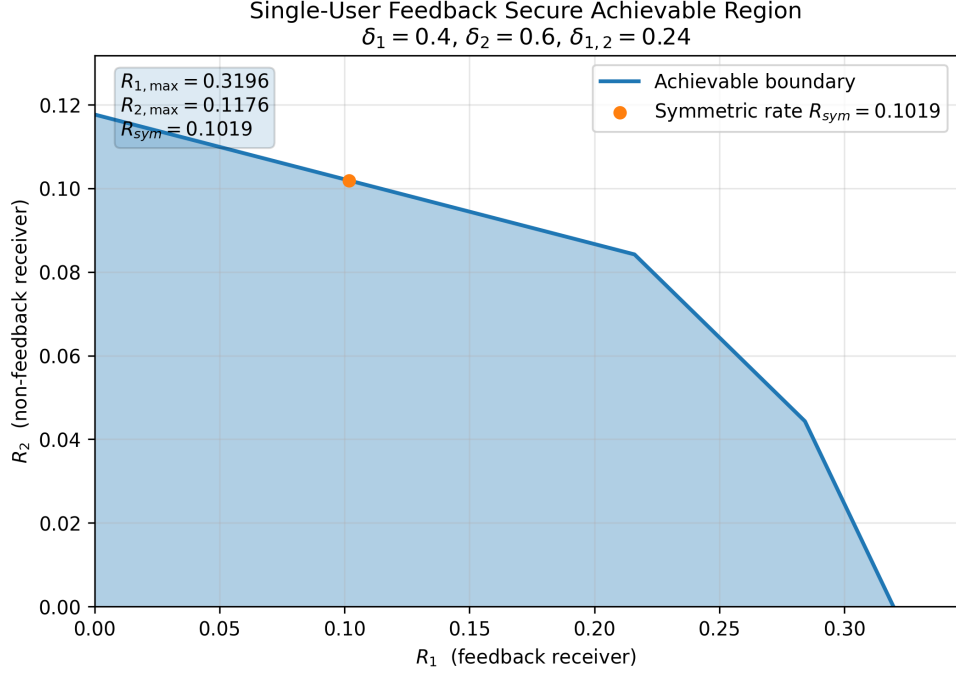


Figure 1: Achievable secure rate region of the two-user BPEC with single-user feedback for $\delta_1 = 0.4$, $\delta_2 = 0.6$, and $\delta_{1,2} = 0.24$.

Fig. 1 shows that the achievable secure rate region is nonzero even though only R_{X_1} provides feedback. The numerical intercepts are

$$R_{1,\max} = 0.3196, \quad R_{2,\max} = 0.1176, \quad R_{sym} = 0.1019,$$

where $R_{2,\max}$ denotes the maximum individual secure rate of R_{X_2} when $R_1 = 0$, and R_{sym} denotes the maximum symmetric rate satisfying $R_1 = R_2$.

The key observation is the positive value of $R_{2,\max}$. Since R_{X_2} does not provide feedback and also has a larger erasure probability than R_{X_1} , this positive value indicates that the non-feedback receiver can still benefit from the transmission structure generated by R_{X_1} 's feedback. The positive symmetric rate further shows that both receivers can simultaneously achieve nonzero secure rates under this channel setting. This numerical result is consistent with the interpretation in Section 3.2, the feedback from R_{X_1} helps create useful randomization and side information, which indirectly supports secure communication for R_{X_2} .

4.2 Comparison with Baseline Regions

According to Fig 2, the full-feedback secure capacity region is used as an upper benchmark, since both receivers provide channel state feedback in that setting. The no-feedback secure region is used as a lower

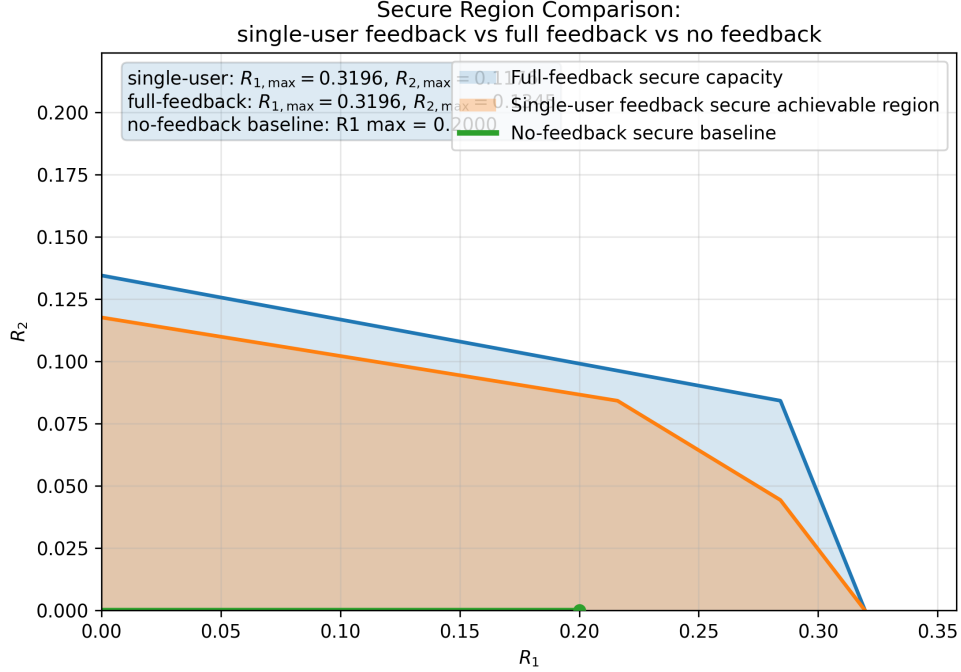


Figure 2: Comparison among the single-user feedback secure achievable region, the full-feedback secure capacity region, and the no-feedback secure baseline for $\delta_1 = 0.4$, $\delta_2 = 0.6$, and $\delta_{1,2} = 0.24$.

baseline, since neither receiver provides state feedback. These two baselines help quantify how much secrecy-rate gain is provided by single-user feedback.

Based on the same channel parameters assumptions, the single-user feedback achievable region is compared with several baseline regions. In this case, Rx₂ is the weaker receiver because $\delta_2 > \delta_1$. Therefore, without feedback, Rx₂ has zero individual secure rate. The comparison highlights that the feedback from Rx₁ can still support secure communication for the non-feedback receiver.

Fig. 2 shows that the full-feedback secure capacity region is the largest among the secure regions, as expected. The single-user feedback achievable region is smaller than the full-feedback secure region, but it remains much larger than the no-feedback secure baseline. The no-feedback secure baseline gives a positive individual secure rate only for Rx₁, while Rx₂ has zero individual secure rate.

The main observation is that the loss from removing Rx₂'s feedback is visible but not catastrophic. The maximum secure rate of Rx₂ decreases from 0.1345 under full feedback to 0.1176 under single-user feedback. However, this value is still strictly positive, whereas the no-feedback baseline gives zero rate for Rx₂. This result shows that single-user feedback does not fully recover the full-feedback secure capacity region, but it still provides a substantial secrecy-rate gain for the non-feedback receiver.

Moreover, the second comparison focuses on the cost of imposing secrecy constraints. Here, the full-feedback secure capacity region is compared with the non-secure feedback capacity region of the two-user

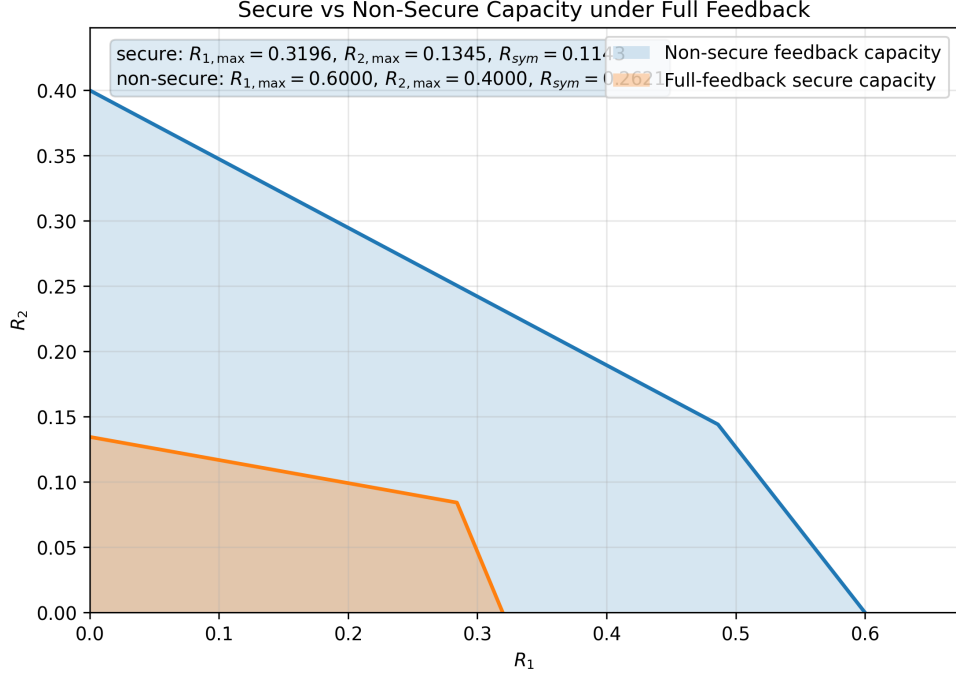


Figure 3: Comparison between the full-feedback secure capacity region and the non-secure feedback capacity region for $\delta_1 = 0.4$, $\delta_2 = 0.6$, and $\delta_{1,2} = 0.24$.

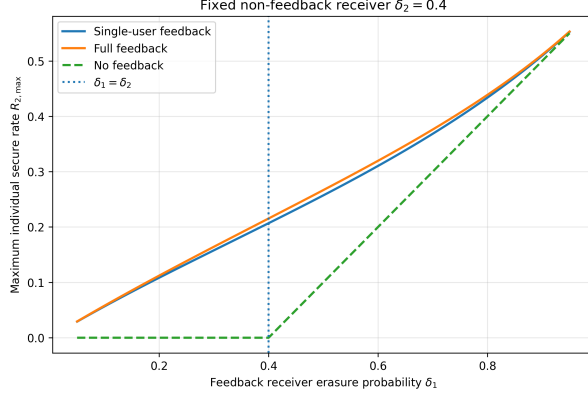
BPEC. The non-secure region only requires reliable decoding, while the secure region also requires that each receiver's message remain information-theoretically secret from the other receiver. Therefore, the gap between the two regions can be interpreted as the rate cost of secrecy.

Fig. 3 shows that the secure capacity region is significantly smaller than the non-secure feedback capacity region. Secrecy changes the shape of the achievable region because part of the communication resource must be used to create or consume secrecy resources, such as secret keys and randomization. Thus, the secure capacity region can be viewed as the non-secure feedback transmission structure together with an additional secrecy overhead.

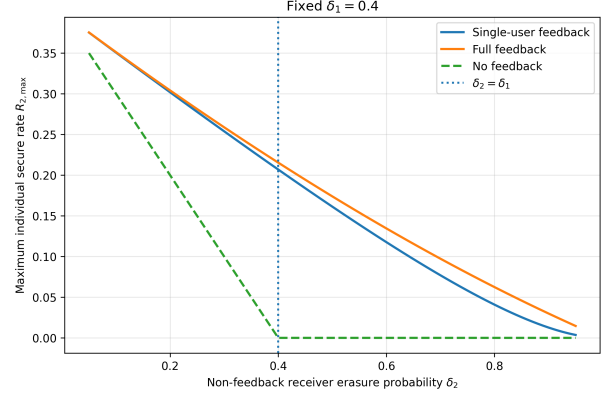
4.3 Parameter Sweep: Effect of Erasure Probability

This subsection further studies how the achievable secure rate changes with the erasure probabilities. Therefore, the channel parameters is varied to examine how the relative channel strengths of Rx₁ and Rx₂ affect the secure rate of the non-feedback receiver. The main quantity of interest is $R_{2,\max}$, which denotes the maximum individual secure rate of the non-feedback receiver when $R_1 = 0$.

First, consider a one-dimensional sweep by fixing the erasure probability of the non-feedback receiver as $\delta_2 = 0.5$, while varying the erasure probability of the feedback receiver, δ_1 . Under the independent erasure assumption. This sweep demonstrates how the quality of the feedback receiver's channel affects the



(a) Fixed $\delta_2 = 0.4$, varying δ_1 .



(b) Fixed $\delta_1 = 0.4$, varying δ_2 .

Figure 4: Parameter sweeps of the maximum individual secure rate $R_{2,\max}$ under single-user feedback, full feedback, and no feedback baselines.

achievable secure rate of the non-feedback receiver.

Next, consider the complementary sweep. The feedback receiver's erasure probability is fixed as $\delta_1 = 0.4$, while the non-feedback receiver's erasure probability δ_2 is varied. This sweep shows the effect of the non-feedback receiver becomes weaker.

Fig. 4 compares the maximum individual secure rate of Rx₂ under single-user feedback, full feedback, and no feedback. In Fig. 4a, when $\delta_1 < \delta_2$, Rx₁ has the stronger channel and Rx₂ is the weaker receiver. In this region, the no-feedback baseline gives zero secure rate for Rx₂. However, the single-user feedback curve remains positive over a nontrivial range. This shows that the feedback from Rx₁ can still help Rx₂ achieve a positive secure communication rate even though Rx₂ itself provides no feedback.

When $\delta_1 > \delta_2$, Rx₂ has the stronger channel. In this case, the no-feedback baseline becomes positive because Rx₂ already has a channel advantage over Rx₁. Nevertheless, the single-user feedback curve remains above the no-feedback baseline. This indicates that single-user feedback does more than simply recover the ordinary wiretap advantage. It can further improve the achievable secure rate of the non-feedback receiver by using Rx₁'s feedback to create useful randomization and side information.

Similarly, Fig. 4b shows the same effect from the opposite direction. As δ_2 increases, the non-feedback receiver's channel becomes weaker, so $R_{2,\max}$ generally decreases. The vertical dotted line marks the point $\delta_2 = \delta_1$. To the right of this line, $\delta_2 > \delta_1$, so Rx₂ is the weaker receiver. In this region, the no-feedback secure rate of Rx₂ is zero. However, the single-user feedback curve remains positive for a range of δ_2 values beyond this point. This provides another numerical illustration of the main qualitative conclusion, the non-feedback receiver does not immediately lose all secure communication capability when its channel becomes weaker than the feedback receiver's channel.

Single-user feedback achievable boundary, fixed $\delta_1 = 0.4$

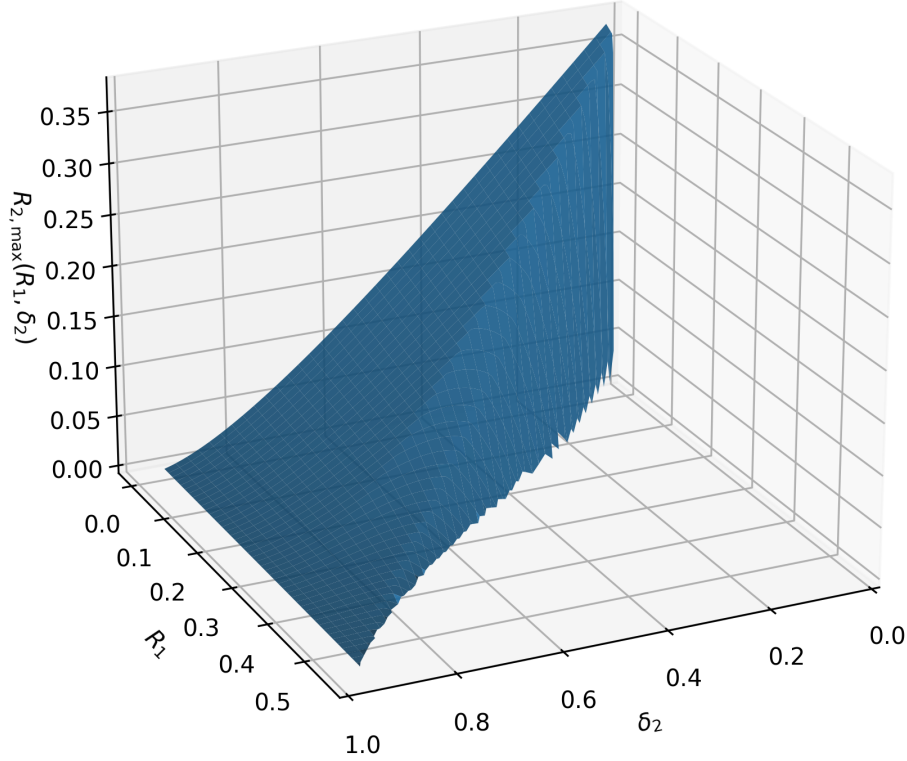


Figure 5: Three-dimensional visualization of the single-user feedback achievable boundary. For each fixed δ_2 , the curve obtained by slicing the surface represents the upper boundary of the achievable region in the (R_1, R_2) plane.

Furthermore, the achievable boundary of the single-user feedback is plotted as a three-dimensional surface. For each value of δ_2 , the program computes the upper boundary

$$R_2 = R_{2,\max}(R_1, \delta_2)$$

of the single-user feedback achievable region. The resulting surface shows how the entire rate tradeoff changes as the non-feedback receiver's erasure probability varies.

Fig. 5 gives a more complete view than the one-dimensional sweeps. As δ_2 increases, the surface moves downward, meaning that the achievable region shrinks as the non-feedback receiver becomes weaker. However, the surface does not immediately collapse to zero after δ_2 exceeds δ_1 . Thus, the three-dimensional plot visually summarizes the same conclusion as the previous sweeps. Single-user feedback can preserve a positive

secure rate for the non-feedback receiver over a meaningful range of weaker-channel conditions.

5 Discussion: Toward General Broadcast Channels

A main difficulty in extending the single-user feedback scheme from the broadcast packet erasure channel to a general broadcast channel is that the original scheme relies heavily on the clear received-or-erased structure of BPECs. In the erasure model, the feedback receiver reports whether each packet is received, so the transmitter can explicitly identify which symbols are known to Rx1 and which symbols are hidden from Rx1. This structure is essential in all three phases of the scheme. The structure supports secret-key generation for the feedback receiver, randomizer construction for the non-feedback receiver, and opportunistic network coding in the final transmission phase. However, in a general noisy broadcast channel, the receiver output is not simply a packet or an erasure. Instead, each receiver observes a noisy version of the transmitted signal, and the transmitter cannot directly classify symbols as “known” or “unknown” at a receiver. As a result, the key-generation and side-information structure becomes much less explicit.

Another challenge is that the Phase 3 network-coding argument depends on the transmitter knowing which packets were received by Rx1 and then retransmitting linear combinations that do not create additional secrecy leakage. This works naturally in BPECs because side information is discrete and identifiable. In a general channel, receiver observations are soft, noisy, and statistically correlated rather than clean packet subsets. Therefore, it is unclear how to define an equivalent randomizer, how to measure the useful side information at the non-feedback receiver, and how to guarantee that retransmission does not leak extra information to the feedback receiver. Therefore, extending single-user feedback to general broadcast channels likely requires a new coding framework rather than a direct modification of the three-phase BPEC scheme.

6 Conclusion

This report reviewed secure communication over broadcast erasure channels with feedback, with emphasis on how feedback changes the achievable secrecy performance in multi-user networks. Starting from wiretap channel by Wyner, secrecy was introduced as an information-theoretic notion measured through equivocation and confidential message rates. This classical framework provides the foundation for understanding why degraded receivers, eavesdropper observations, and channel asymmetry are central factors in secure communication design.

The discussion then moved to broadcast packet erasure channels, where feedback has a particularly explicit operational meaning. Unlike general noisy broadcast channels, the transmitter can directly learn

which packets were successfully received and which packets were erased. This property allows feedback to be transformed into retransmission opportunities, coded packet mixing, and secret-key generation based on asymmetric packet receptions. As a result, the secure rate region can be significantly enlarged compared with the no-feedback setting.

The main focus of the report was the single-user feedback model. Existing results show that even when only one receiver provides channel-state feedback, positive secrecy gains are still achievable for both users under suitable channel conditions. This demonstrates that full feedback from every receiver is not always necessary to obtain substantial security benefits. Carefully designed multi-phase schemes can exploit partial feedback for key generation, wiretap coding randomization, and opportunistic network coding simultaneously.

Simulation results were included to illustrate how erasure probabilities affect the achievable secure rates. The numerical trends are consistent with the theoretical expressions: stronger legitimate links and more favorable feedback structures improve the secure rate region, while harsher erasure conditions increase the secrecy overhead.

Overall, the literature indicates that feedback is a powerful resource for secure broadcast communication. Future work may investigate extensions from erasure models to more general noisy broadcast channels, robustness under delayed or imperfect feedback, and practical finite-blocklength coding schemes that bridge the gap between theoretical secrecy capacity and real wireless systems.

A Simulation Code

<https://github.com/Stanlet-L/ECE679/releases/tag/ECE679>

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